

E-COMMERCE RETURN ANALYSIS

1. Introduction

E-commerce businesses handle a large number of orders, and product returns can affect operational efficiency, customer satisfaction, business costs, and sustainability. Understanding the patterns and reasons behind returns helps businesses identify problem areas and make better decisions.

The E-Commerce Return Analysis project analyzes transaction data to identify return patterns across product categories, return reasons, discounts, locations, payment methods, shipping methods, and order values. It also examines the financial and environmental impact of returns. The project uses Python, SQL, PostgreSQL, and Power BI to transform transaction data into meaningful business insights through an interactive dashboard.

2. Abstract

The E-Commerce Return Analysis project is an end-to-end data analytics project developed to understand product returns in an e-commerce environment. The dataset contains 5,000 transaction records covering orders, products, customers, discounts, payment and shipping methods, return details, costs, and environmental metrics.

The project involved data cleaning and exploratory analysis using Python and Pandas, followed by structured analysis using SQL and PostgreSQL. Finally, Power BI and DAX were used to develop an interactive dashboard for visual analysis and reporting.

The analysis identified an overall return rate of 29%, with 1,450 returned orders out of 5,000. Clothing recorded the highest return rate at 37.43%, while Defective products were the most common return reason. Return patterns were also analyzed across discounts, locations, payment methods, and shipping methods. The total recorded return cost was ₹290,000.

The project demonstrates how data analytics can convert raw transaction data into useful business insights and support data-driven decision-making.

3. Tools Used

- **Python & Pandas** – Data cleaning, analysis, and exploratory data analysis
- **Matplotlib** – Data visualization
- **PostgreSQL & SQL** – Data storage and structured analysis
- **Power BI & DAX** – Interactive dashboard and dynamic KPI calculations
- **GitHub** – Project documentation and version management

4. Steps Involved in Building the Project

Step 1: Dataset Selection

A relevant e-commerce returns dataset containing 5,000 records was selected, covering order, product, customer, return, financial, and environmental information.

Step 2: Data Cleaning

The dataset was examined and prepared for analysis by checking data types, missing values, duplicate records, categorical values, and numerical fields.

Step 3: Exploratory Data Analysis

Python and Pandas were used to analyze return rates, product categories, return reasons, high-return products and locations, discounts, order values, payment methods, shipping methods, financial impact, and environmental impact.

Step 4: SQL and PostgreSQL Analysis

The cleaned dataset was imported into PostgreSQL. SQL queries were used to calculate return rates, analyze return reasons, identify high-risk products and locations, and examine financial and sustainability metrics.

Step 5: Power BI Dashboard

The PostgreSQL data was connected to Power BI to create an interactive dashboard containing:

- Total Orders, Returned Orders, Return Rate, and Total Return Cost
- Returns by Product Category and Return Reasons
- Discount and Order Value analysis
- Top 10 Locations by Orders
- Return Cost by Category
- Environmental Impact
- Payment and Shipping Method analysis

Interactive slicers were added for Return Status, Payment Method, and Product Category.

Step 6: Insight Generation

The results were analyzed to identify important business patterns and potential improvement areas, including high-return categories and products, return-related operational issues, and opportunities to reduce unnecessary returns.

5. Conclusion

The E-Commerce Return Analysis project successfully demonstrates an end-to-end data analytics workflow, from data preparation and exploratory analysis to SQL-based analysis and interactive Power BI visualization.

The analysis found a 29% overall return rate, with Clothing having the highest return rate at 37.43%. Defective products were the most common return reason, while the total recorded return cost was ₹290,000. The project also identified variations in return patterns across discounts, payment methods, locations, and product categories.

The Power BI dashboard provides an interactive way to understand these patterns and supports better decision-making. Overall, the project demonstrates the practical use of Python, SQL, PostgreSQL, Power BI, and DAX to transform e-commerce data into actionable business insights.